In Fig, 6.8 (a) two rows are fixed. We get only one possible way to complete the third row (Fig. 6.8 (b)).

(ii) 3X3 sudoku

1	2	3

Fig. 6.9

In the above 3 x 3 sudoku, the first row only is fixed. The second row can be filled in 2 ways either by 2 3 1 or 3 1 2

1	2	3
2	3	1
3	1	2

(a)

1	2	3
3	1	2
2	3	1

(b)

Fig. 6.10

To fill the third row, bear in mind that numbers cannot be repeated in the row or in the column. The third row can be filled by only one way in each case.

(iii) 3 × 3 sudoku

In how many different ways can the first row be arranged?

The first row can be arranged in six ways as (1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2) and (3,2,1).

(iv) Let us find all possible solutions to solve 3 × 3 sudoku puzzle.

1	2	3
2	3	1
3	1	2

(a)

1	2	3
3	1	2
2	3	1

(b)

2	3	1
3	1	2
1	2	3

(c)

2	3	1
1	2	3
3	1	2

(d)

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